



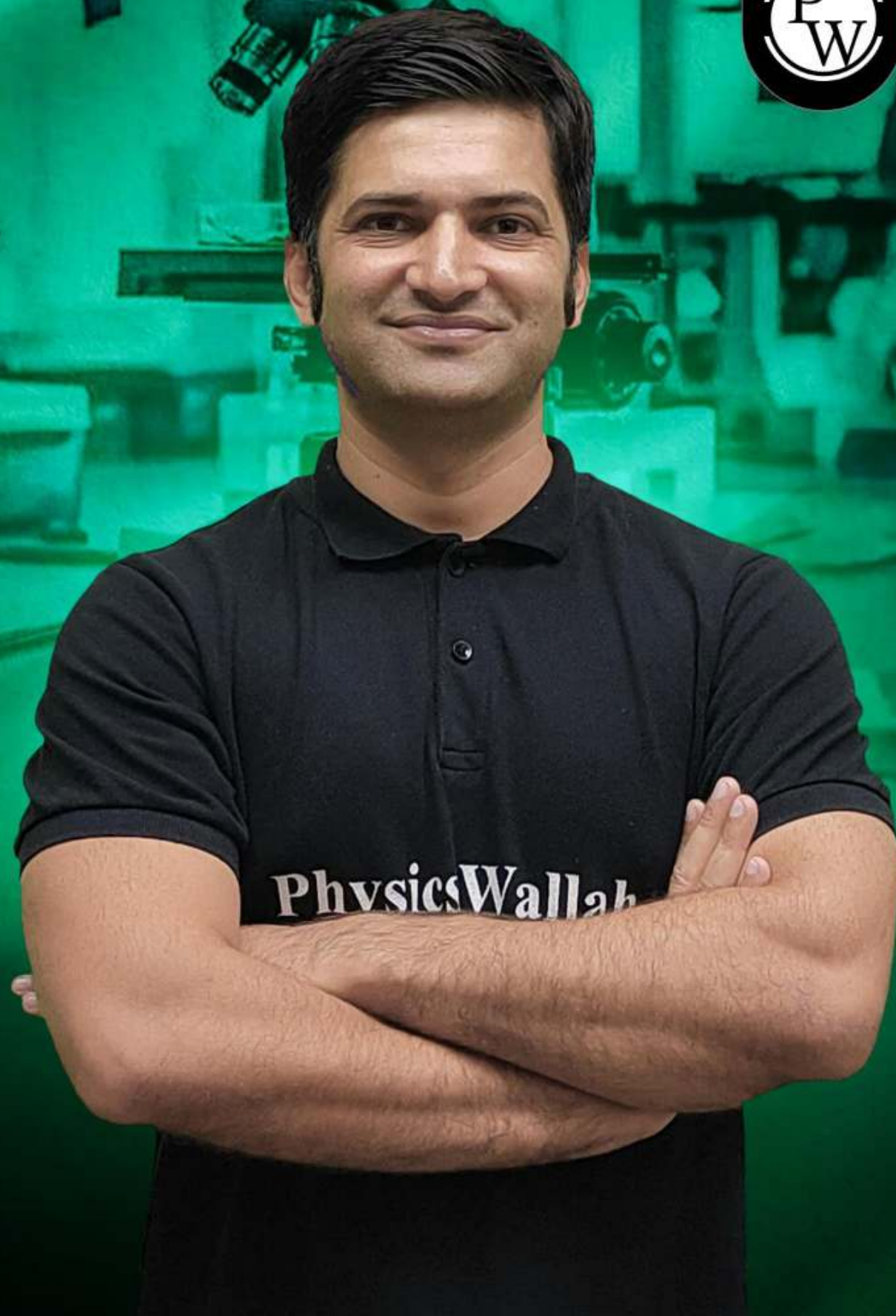
NEET 2.0 2027

Botany

Lecture - 08

Cell : The Unit of Life

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Topics to be covered

1

Lysosome

2

vacuole

3

Mitochondria

4

Golgi Body

★ Camillo golgi, 1898

★ dense stain, Reticular (NET)
near nucleus

★ Flat, disc/sac like: Cisternae

★ Plant: 'Dictyosome' use
for golgi Body

stack

parallel

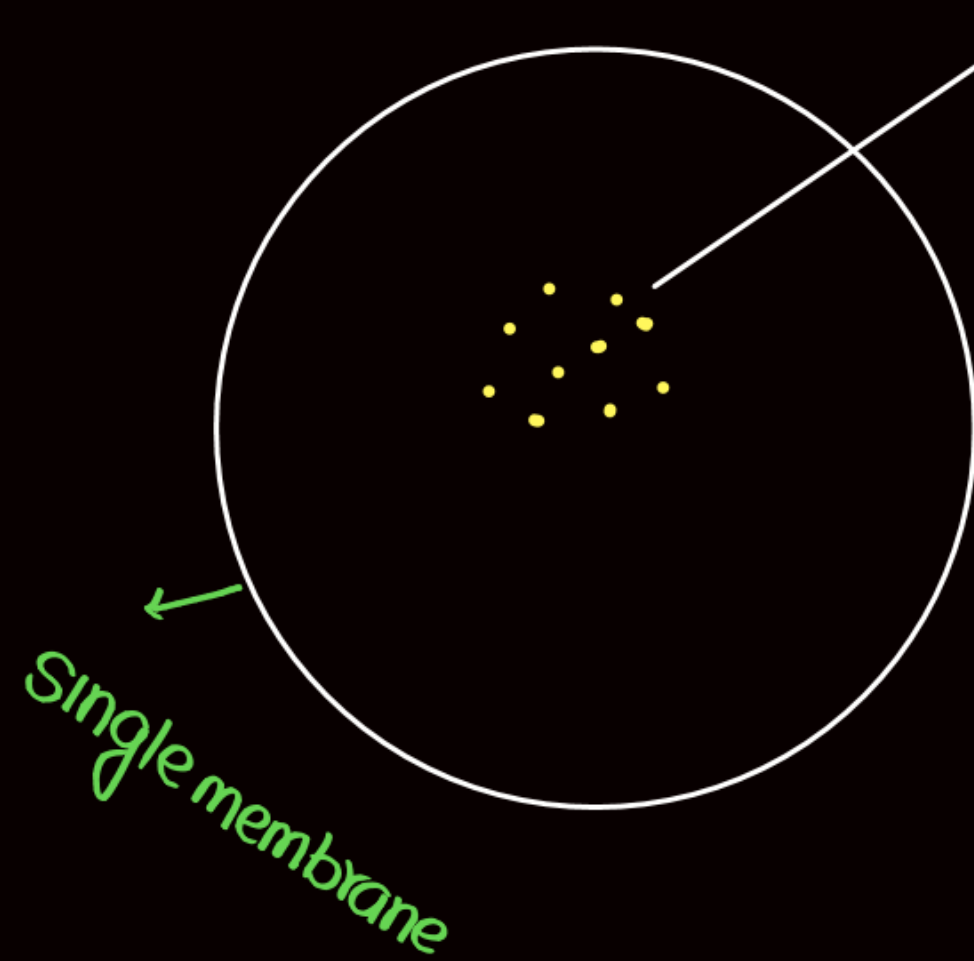
0.5-1 μ m (diameter)

Concentrically arrange

No vary in golgi Body.

NOTE: Sarcoplasmic Reticulum [SER]
SKIP store Ca^{+2} , muscle contraction
detoxification of drug: SER.

Lysosome



digestive enzyme
- (Food digestion)
OR
HYDROLYTIC ENZYME
(Need H_2O to perform
Function)

ACID HYDROLASES

WORK IN
ACIDIC
MEDIUM
(PH: 5)

→ formed By golgi Body due to packaging

→ Carbohydrases
proteases
lipase
Nuclease

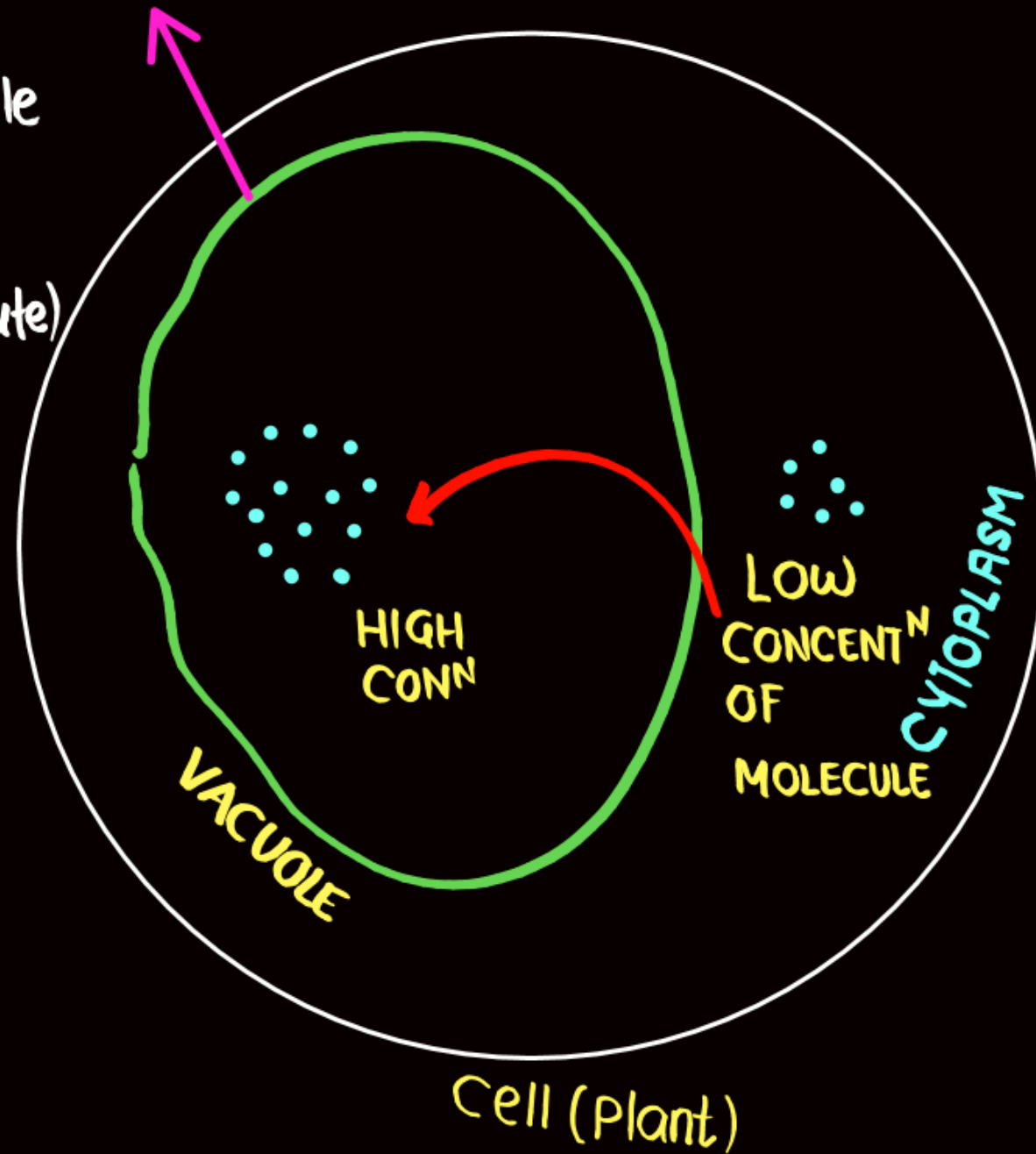
NOTE : Polymorphic organelle.
↓
more than one Type.

Vacuole

★ **membrane:** Tonoplast/single (selectively permeable) (allow some solute)

★ **Large vacuole:** 90% volume in cell.

★ **molecule move from** cytoplasm to vacuole [Low to high / against concⁿ gradient / Active Transport]



★ membrane Bound space in cytoplasm.

★ H_2O , salt (sap), excretory material & material not useful for cell.

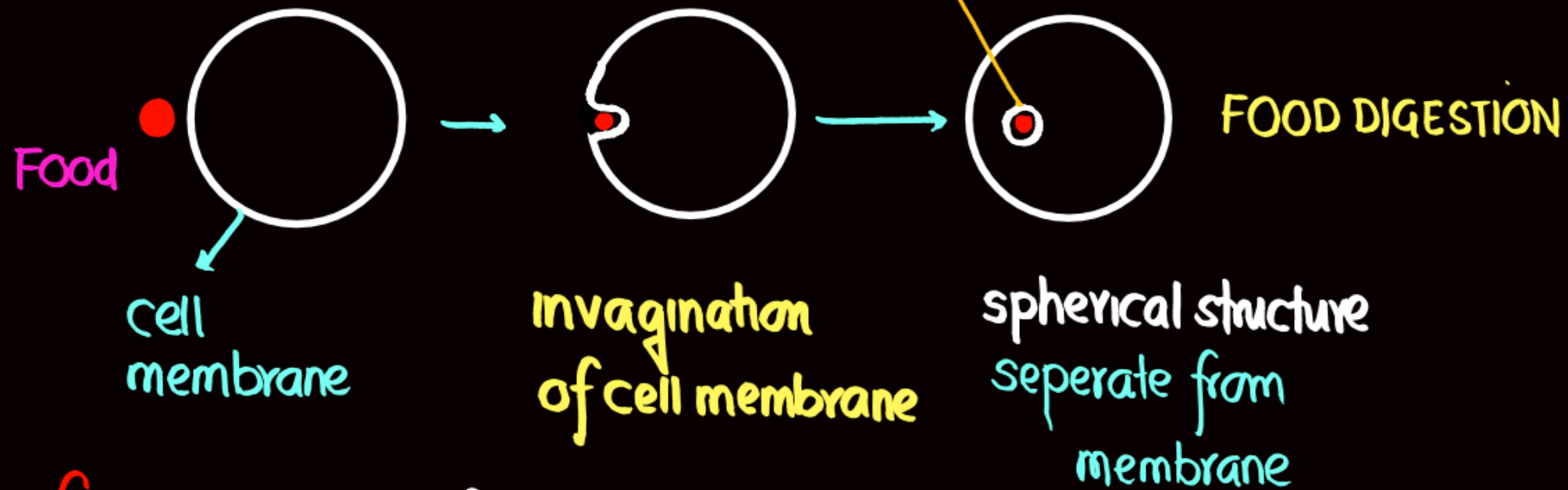
Note: Contractile vacuole

★ Amoeba, paramecium

★ Remove excess water from Body: Osmoregulation & Remove waste (Excretion)

Food vacuole FV

⇒ protista (paramecium)



⇒ **formed** by engulfing of food through cell membrane

MITOCHONDRIA

- ★ double membrane

- ★ not visible until stain by
Dye

- ★ sausage/cylindrical shape

- ★ number depend upon

Physiology of cell

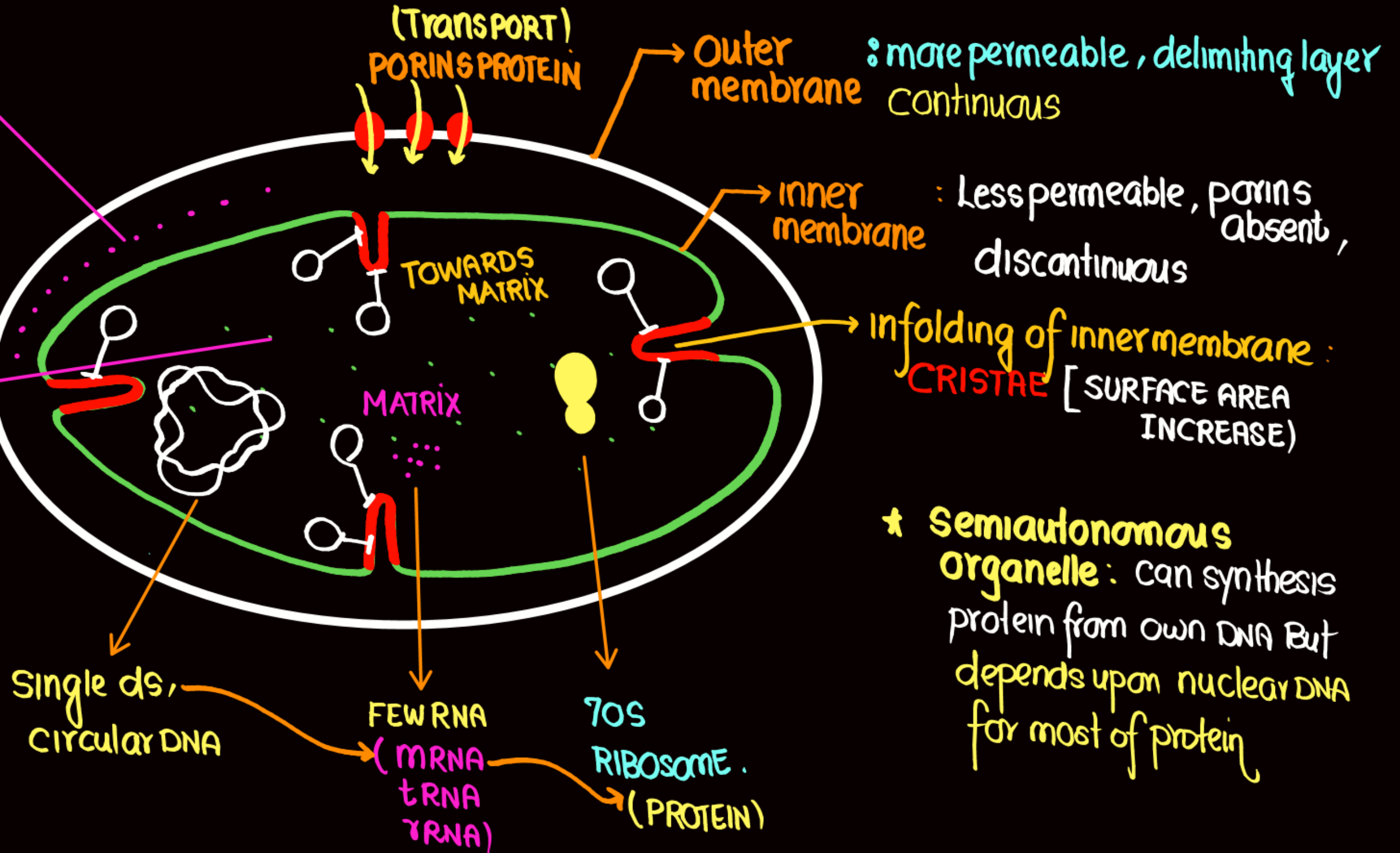
germinating seed: } MORE
Flight muscle (BIRD) } MITOCHONDRIA

SPACE B/w OUTER
AND INNER MEMBRANE

INTERMEMBRANOUS
SPACE / OUTER
COMPARTMENT

INNER COMPARTMENT

[Filled with
homogenous
substance:
MATRIX

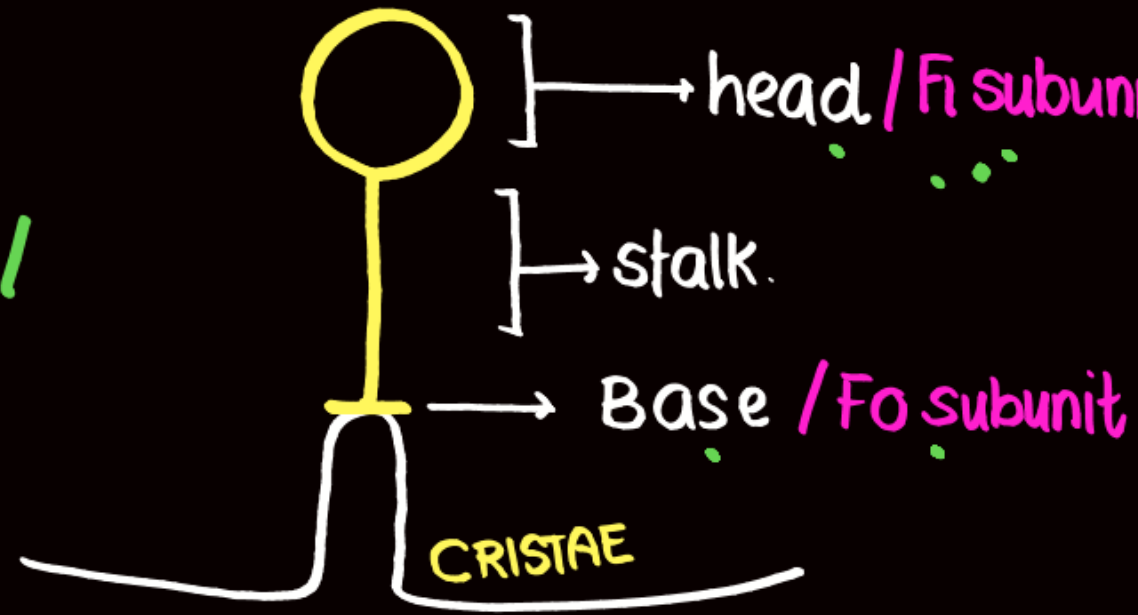


KHUSHI

SPAMMING
की.

SUBSCRIPTION
CANCEL करना
है

- * F₁F₀ particle / elementary particle
- * Oxysome
- * ATP synthase



main site [ATP FORMATION]



[OXIDATIVE PHOSPHORYLATION] ✓

- * POWER HOUSE OF CELL
- * Aerobic Respiration

8.5.3.4 Vacuoles

The vacuole is the ^{inorganic salt} membrane-bound space found in the cytoplasm. It contains water, sap, excretory product and other materials not useful for the cell. The vacuole is bound by a single membrane called tonoplast. In plant cells the vacuoles can occupy up to 90 per cent of the volume of the cell.

Question

Vacuole

- ☒ (A) ~~double membrane bound~~ space in cytoplasm
- ☒ (B) It contain water, sap, excretory product
- ☒ (C) ~~it's membrane is tonoplast which is semipermeable~~
- ☐ (D) occupy 90% volume in ~~animal~~ cell
- ☐ (E) all are incorrect

In plants, the tonoplast facilitates the transport of a number of ions and other materials against concentration gradients ^{from cytoplasm.} into the vacuole, hence their concentration is significantly higher in the vacuole than in the cytoplasm. ↓

Question

- ☒ **Assertion (A):** material enter into vacuole from cytoplasm by active transport
- ☒ **Reason (R):** material move against concentration gradient into vacuole
- ☒ (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

In *Amoeba*, the contractile vacuole is important for osmoregulation and excretion. In many cells, as in protists, food vacuoles are formed by engulfing the food particles.

Question

Correct statement

- (A) Contractile vacuole found only in ~~amoeba~~ *param.*
- (B) contractile vacuole perform only excretion ~~not~~ osmoregulation
- ☒ (C) In Protista food vacuole present formed by engulfing food
- (D) none

8.5.3.3 Lysosomes

vesicle like ○



These are membrane bound vesicular structures formed by the process of packaging in the golgi apparatus. The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes (hydrolases – lipases, proteases, carbohydrases) optimally active at the acidic pH. These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids. (Nucleases)

Question

Lysosome

- A. ~~double~~ membrane
- B. former by packaging in ER ~~ER~~
- C. rich in hydrolytic enzymes work in ~~basic~~ medium
- ☒ D. enzymes work in acidic medium PH-5
- E. enzymes digest carbohydrate protein only ~~only~~

(A) 0 ☒ (B) 1 (C) 2 (D) 3



8.5.4 Mitochondria

Mitochondria (sing.: mitochondrion), unless specifically stained, are not easily visible under the microscope. The number of mitochondria per cell is variable depending on the physiological activity of the cells. In terms of shape and size also, considerable degree of variability is observed. Typically it is sausage-shaped or cylindrical having a diameter of 0.2-1.0 μ m (average 0.5 μ m) and length 1.0-4.1 μ m.

Question

Correct statement

- A. Mitochondria not visible even after staining
- B. number of mitochondria is not constant
- C. it's number depend upon physiology
- D. shape, size is constant in mitochondria
- E. it is sausage shape
- F. diameter 0.2 to 0.1 μ m length 1-4.1 μ m

(A) 1 (B) 2 (C) 3 (D) 4

Question

Mitochondria

- (A) single membrane organelle
- (B) energy currency of cell
- (C) ATP factory of cell
- (D) both (B) & (C) are correct

Each mitochondrion is a double membrane-bound structure with the outer membrane and the inner membrane dividing its lumen distinctly into two aqueous compartments, i.e., the outer compartment and the inner compartment. The inner compartment is filled with a dense homogeneous substance called the matrix. The outer membrane forms the continuous limiting boundary of the organelle.

BONUS

Question

Outer: IMS
Mitochondria (Incorrect statement)

- ☒ A. membrane divide its lumen into two compartment *matrix*
- ☒ B. inner compartment filled with ~~heterogenous~~ substance matrix
- ☒ C. outer membrane is delimiting layer and ~~discontinuous~~
- ☒ D. Inner membrane forms infolding called ~~cisternae~~ **CRISTAE**

(A) 1 (B) 2 ☒ (C) 3 (D) 4

Question

Correct statement

- (A) Cristae form ~~away~~ *Towards* from matrix
- (B) cristae ~~decrease~~ *incree* surface area
- ☒ (C) inner membrane is discontinuous
- (D) all are incorrect

8.5.3.2 Golgi apparatus

Camillo Golgi (1898) first observed densely stained reticular structures near the nucleus. These were later named Golgi bodies after him. They consist of many flat, disc-shaped sacs or cisternae of 0.5µm to 1.0µm diameter (Figure 8.6). These are stacked parallel to each other. Varied number of cisternae are present in a Golgi complex.

NET LIKE



Question



Correct statement

- ☒ A. Camilo Golgi (1808) observed lightly stain structure
- ☒ B. It present near nucleus
- ☒ C. GB consist of flat, disc like cisternae which are parallel to each other
- ☒ D. cisternae number fixed in GB
- ☒ E. Cisternae concentrically arranged near mitochondria

(A) 2 (B) 2 (C) 3 (D) 4

BONUS

Question

Incorrect statement for Golgi body

- ☒ (A) Golgi cisternae arrange near nucleus with cis / forming side face and trans/maturing face
- ☒ (B) Both face are interconnected and different, perform same function
- ☒ (C) formation of glycoprotein and Glycolipid
- ☒ (D) Golgi body is not closely associated with ER
- ☒ (E) both (B) and (D)



Homework From **Yakeen NEET Module**



Exercise – 01

Q. 65, 66, 69, 70, 75, 80, 84, 85

Thank

You